Practical Week-10

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samanth-ubuntu@samanth-ubuntu:~$ echo "Hello Linux Inode" > original.txt
samanth-ubuntu@samanth-ubuntu:~$ cat original.txt
Hello Linux Inode
samanth-ubuntu@samanth-ubuntu:~$ ls -i original.txt
42849 original.txt
samanth-ubuntu@samanth-ubuntu:~$ 42849 → inode number
original.txt → file nam
42849: command not found
original.txt: command not found
samanth-ubuntu@samanth-ubuntu:~$ stat original.txt
  File: original.txt
  Size: 18                Blocks: 8          IO Block: 4096   regular file
Device: 8,48            Inode: 42849      Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ samanth-ubuntu)   Gid: ( 1000/ samanth-ubuntu)
Access: 2026-10-04 16:02:10.056018883 +0000
Modify: 2026-10-04 16:02:01.796020472 +0000
Change: 2026-10-04 16:02:01.796020472 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
samanth-ubuntu@samanth-ubuntu:~$ ln original.txt hardlink.txt
samanth-ubuntu@samanth-ubuntu:~$ ls -li original.txt hardlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  18 Oct  4 16:02  hardlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  18 Oct  4 16:02  original.txt
samanth-ubuntu@samanth-ubuntu:~$ echo "Modified through hard link" > hardlink.txt
samanth-ubuntu@samanth-ubuntu:~$ cat original.txt
Modified through hard link
samanth-ubuntu@samanth-ubuntu:~$ ln -s original.txt symlink.txt
samanth-ubuntu@samanth-ubuntu:~$ ls -li original.txt hardlink.txt symlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  27 Oct  4 16:06  hardlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  27 Oct  4 16:06  original.txt
42850 lrwxrwxrwx 1 samanth-ubuntu samanth-ubuntu  12 Oct  4 16:06  symlink.txt -> original.txt
samanth-ubuntu@samanth-ubuntu:~$ stat original.txt
  File: original.txt
  Size: 27                Blocks: 8          IO Block: 4096   regular file
Device: 8,48            Inode: 42849      Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ samanth-ubuntu)   Gid: ( 1000/ samanth-ubuntu)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
samanth-ubuntu@samanth-ubuntu:~$ stat srigLink.txt
  File: original.txt
  Size: 27                Blocks: 8          IO Block: 4096   regular file
Device: 8,48            Inode: 42849      Links: 2
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Change: 2026-10-04 16:06:13.643975194 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
samanth-ubuntu@samanth-ubuntu:~$ stat symlink.txt
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samanth-ubuntu@samanth-ubuntu:~$ ls -l symLink.txt
  File: 'symlink.txt' -> 'original.txt'
  Size: 12              Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 42850       Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ samanth-ubuntu)   Gid: ( 1000/ samanth-ubuntu)
Access: 2026-10-04 16:06:39.135970423 +0000
Modify: 2026-10-04 16:06:31.111971931 +0000
Change: 2026-10-04 16:06:31.111971931 +0000
 Birth: 2026-10-04 16:06:31.111971931 +0000
samanth-ubuntu@samanth-ubuntu:~$ stat -L symlink.txt
  File: symlink.txt
  Size: 27              Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 42849       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ samanth-ubuntu)   Gid: ( 1000/ samanth-ubuntu)
Access: 2026-10-04 16:06:21.159973809 +0000
Modify: 2026-10-04 16:06:13.643975194 +0000
Change: 2026-10-04 16:06:13.643975194 +0000
 Birth: 2026-10-04 16:02:01.792020472 +0000
samanth-ubuntu@samanth-ubuntu:~$ ls -i original.txt
42849 original.txt
samanth-ubuntu@samanth-ubuntu:~$ find . -inum 123456
samanth-ubuntu@samanth-ubuntu:~$ ls -li original.txt hardlink.txt symlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  27 Oct  4 16:06 hardlink.txt
42849 -rw-r--r-- 2 samanth-ubuntu samanth-ubuntu  27 Oct  4 16:06 original.txt
42850 lrwxrwxrwx 1 samanth-ubuntu samanth-ubuntu  12 Oct  4 16:06 symlink.txt -> original.txt
samanth-ubuntu@samanth-ubuntu:~$ stat original.txt
  File: original.txt
  Size: 27              Blocks: 8          IO Block: 4096   regular file
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  File: hardlink.txt
  Size: 27              Blocks: 8          IO Block: 4096   regular file
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  Size: 12              Blocks: 0          IO Block: 4096   symbolic link
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Modify: 2026-10-04 16:06:31.111971931 +0000
```

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  File: original.txt
  Size: 27                Blocks: 8          IO Block: 4096   regular file
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Device: 8,48     Inode: 42850         Links: 1
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 Birth: 2026-10-04 16:06:31.111971931 +0000
samanth-ubuntu@samanth-ubuntu:~$ nano mmap_file.c
samanth-ubuntu@samanth-ubuntu:~$ gcc mmap_file.c -o mmap_file
samanth-ubuntu@samanth-ubuntu:~$ ./mmap_file
Data read using mmap:
Hello from mmap()!
File written successfully using mmap().
samanth-ubuntu@samanth-ubuntu:~$ cat mmap_test.txt
Hello from mmap()!
samanth-ubuntu@samanth-ubuntu:~$
```